HRI Study Design

David Rusia, Saba Khaburzania, Yassin Baher, Ana Saldadze

Research Question:
Does the source of a hint (a physical robot arm using motion cues, or a screen) change how much people rely on that hint while physically sorting blocks, and how their reliance and trust change after incorrect hints and as the task gets harder?

[Optional] More Specific Research Question(s):

1. Does a physical robot arm using motion cues lead to greater reliance on its hints than an equivalent screen-based hint?
2. How does an incorrect hint affect subsequent reliance on hints in the robot condition compared with the screen condition?
3. Does increasing task difficulty (easy, medium, hard stages) increase participants' reliance on the hints?
4. After an error, does reliance recover differently when the hint comes from the physical robot rather than from the screen?

Constructs You Intend to Investigate:
Hoffman and Zhao (2020) Section 2.2 "Identify Your Constructs"

- Hint source (embodiment): whether the hint is delivered by the physical robot arm through motion cues or by an equivalent screen-based system. The sorting task itself is always physical.
- Human reliance: how often participants follow the hint when choosing which bowl to put the block into.
- Trust: participants' perceived trust in the robot and in the screen as hint sources, rated separately at the end of the session.
- Trust recovery after errors: how reliance changes on the trials after an incorrect hint, and whether it recovers afterward.
- Cognitive load: increasing task difficulty across the easy, medium and hard stages.

Hypotheses:
Hoffman and Zhao (2020) Sections 3.1 - 3.3 "Hypotheses"

- **H1:** Participants will follow hints delivered by the physical robot (motion cues) more often, and rate the robot higher in trust, than equivalent hints delivered through the screen.
- **H2:** An incorrect hint will reduce subsequent reliance on the hints, measured on the trial after the error.

- **H3:** Reliance after an error will drop and recover differently for robot hints than for screen hints.
- **H4:** As the task becomes more difficult (easy → medium → hard), participants will rely more on the hints.

Experimental Conditions:
Hoffman and Zhao (2020) Section 4.3 "Operationalizing Constructions into Variables and Measures"

- **Condition 1: Robot Hint (Motion Cue)**
  The robot places a block on the shared workspace and gives a hint about which bowl it belongs in through a motion cue. The participant then physically places the block in a bowl of their choice.

- **Condition 2: Screen Hint**
  The robot places the block in the same way, but the hint is shown on a screen instead of being given through robot motion. The participant again physically places the block themselves.

- The **hint content, hint accuracy, task difficulty, timing, and pattern of correct/incorrect hints should be kept equivalent across both sources** so that the main difference is the form of the hint. The robot's motion cue will be pretested to make sure it points to one specific bowl as clearly as the screen hint, and the screen hint timing will be matched to the robot cue.

Experimental Design:
Hoffman and Zhao (2020) Section 4.2 "Between- and Within- Participants Designs"

Will your study be between-subjects or within-subjects?

Within-subjects design.
Each participant will experience both hint sources within the same session:
- **Robot hints** (motion cues)
- **Screen hints**

This allows us to compare how the **same person** changes their reliance and trust depending on whether the hint comes from the robot or from the screen.

If your study is within-subjects, how will you account for ordering effects?

We will randomize and balance:
- The hint source (robot or screen) is **randomized from round to round**
- Each stage (easy, medium, hard) contains an **equal number of robot and screen hints**

Hint accuracy is set by us in advance and is **identical for both sources**, using a pre-generated sequence balanced across source, stage and hint correctness. About 20–24 trials per stage are planned so that every source/stage combination contains incorrect hints. Because hint sources are interleaved, we record the source and correctness of the previous trial to check for carry-over of an error from one source to the other. The stages run easy → medium → hard, so difficulty is partly confounded with time (fatigue, learning); we note this as a limitation and will counterbalance stage order if it turns out to be feasible.

Measures:
Hoffman and Zhao (2020) Section 4.3.2 "Subjective vs. Objective Measures", 5.3 "Manipulation Check"

Manipulation check:

- Did participants perceive the robot's motion cue as pointing to a specific bowl, as clearly as the screen hint?
- Did participants perceive the medium and hard stages as more difficult than the easy stage?
- Did participants notice that the hints were sometimes incorrect?

Subjective measures:

- Trust in the robot and in the screen as hint sources, rated separately at the end of the session with the same trust scale (rating order counterbalanced)
- Confidence in applying the sorting rules
- Perceived reliability of the robot hints and of the screen hints
- Perceived helpfulness of the hints
- **Perceived physical/social presence** of the robot compared with the screen

Objective measures:

- Followed vs. not followed hints (bowl choice matches the hinted bowl)
- Following incorrect hints as a measure of overreliance, and ignoring correct hints as a measure of underreliance
- Participant sorting accuracy (correct bowl chosen)
- Hint accuracy (set in advance, identical for both sources)
- Decision time before placing the block
- Change in reliance on the trial after an error, by hint source (including carry-over from the other source)
- Change in reliance across the easy, medium and hard stages
- Difference in reliance between robot hints and screen hints

Interview questions:

- Which hint source did you trust more, the physical robot or the screen, and why?
- Did the robot's motion cue affect your decision to follow its hint?
- How did your behavior change after the system made a mistake?
- When the stages became harder, did you feel more dependent on the hints?
- Did you feel differently rejecting a recommendation from the robot compared with rejecting one on the screen?
- What made you decide whether to trust your own judgment of the rule or the system's hint?

Procedure:

Hoffman and Zhao (2020) Section 5.1 "Writing a 'Stage Script'"
In about 2 paragraphs describe the general procedure of the experiment. What will be the task? How will you introduce the task to the participant? How are you going to implement the task? (When) will the participant fill out questionnaires?

*Participants will first receive a short explanation of the sorting rules and complete a few practice trials. In each round, the robot will place a **block with certain characteristics** (color, shape or number) on a shared workspace. The participant is given certain conditions (rules) and must decide **which bowl the block goes into**, then physically places it there. In every round a hint is given by either the **robot (motion cue)** or the **screen**, chosen at random, so the task is always physical and only the source of the hint differs. The session has **three stages of questions about the block (easy, medium, hard)**, and each stage contains the same number of robot hints and screen hints. Hint accuracy is set by us in advance and is the same for both sources; some hints are intentionally incorrect.*

*Hint source is randomized per round using a pre-generated sequence balanced across source, stage and hint correctness, with about 20–24 trials per stage. During the task we record whether the participant followed the hint, decision time, sorting accuracy, hint accuracy, and the change in reliance on the trial after an incorrect hint. **Trust is measured at the end of the session**: participants rate the robot and the screen separately with the same trust scale (rating order counterbalanced), together with perceived reliability, helpfulness and presence, and answer the manipulation-check questions. This is followed by a brief interview about what influenced their decisions.*

Sample Size:

https://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower and Hoffman and Zhao (2020) Section 6

Desired sample size:

Desired sample size: 40 participants
Based on an a priori power analysis for a within-subject comparison of the end-of-session trust ratings (robot vs. screen) with a medium expected effect size (d = 0.5), α = .05, and power = .80, approximately 34 participants are required. We therefore aim for 40 participants to account for potential exclusions or incomplete data. Trial-level reliance (hint source × stage × hint correctness) will be analyzed with a mixed-effects logistic regression with a random intercept per participant; its power will be checked by simulation.

Test family: t tests
Statistical test: Means: Difference between two dependent means (matched pairs)
Type of power analysis: A priori
Effect size dz: 0.5
α err prob: 0.05
Power (1-β): 0.8